Q1. JEE Main 2026 (23 January Shift 1)

Let $f(x)=\int \frac{(2-x^2)\cdot e^x}{(\sqrt{1+x})(1-x)^{3/2}}\,dx$. If $f(0)=0$, then $f\left(\frac{1}{2}\right)$ is equal to:

(1) $\sqrt{3e}-1$ (2) $\sqrt{3e}+1$

(3) $\sqrt{2e}+1$ (4) $\sqrt{2e}-1$

Q2. JEE Main 2026 (28 January Shift 2)

Let $f(x)=\int \frac{dx}{x^{\left(\frac{2}{3}\right)}+2x^{\left(\frac{1}{2}\right)}}$ be such that $f(0)=-26+24\log_e(2)$. If $f(1)=a+b\log_e(3)$, where $a,b\in \mathbf{Z}$, then $a+b$ is equal to :

(1) -26 (2) -11 (3) -5 (4) -18

Q3. JEE Main 2026 (22 January Shift 1)

If $\int(\sin x)^{\frac{-11}{2}}(\cos x)^{\frac{-5}{2}}dx=-\frac{p_1}{q_1}(\cot x)^{\frac{9}{2}}-\frac{p_2}{q_2}(\cot x)^{\frac{5}{2}}-\frac{p_3}{q_3}(\cot x)^{\frac{1}{2}}+\frac{p_4}{q_4}(\cot x)^{\frac{-3}{2}}+C$, where p_i and q_i are positive integers with $\gcd(p_i,q_i)=1$ for $i=1,2,3,4$ and C is the constant of integration, then $\frac{15p_1p_2p_3p_4}{q_1q_2q_3q_4}$ is equal to

Q4. JEE Main 2026 (23 January Shift 2)

Let $I(x)=\int \frac{3dx}{(4x+6)\left(\sqrt{4x^2+8x+3}\right)}$ and $I(0)=\frac{\sqrt{3}}{4}+20$. If $I\left(\frac{1}{2}\right)=\frac{a\sqrt{2}}{b}+c$, where $a,b,c\in N$, $\gcd(a,b)=1$, then $a+b+c$ is equal to

(1) 30 (2) 31 (3) 29 (4) 28

Q5. JEE Main 2026 (24 January Shift 1)

Let $f(t)=\int\left(\frac{1-\sin(\log_e t)}{1-\cos(\log_e t)}\right)dt, t>1$.

If $f\left(e^{\pi/2}\right)=-e^{\pi/2}$ and $f\left(e^{\pi/4}\right)=\alpha e^{\pi/4}$, then α equals

(1) $1+\sqrt{2}$ (2) $-1-\sqrt{2}$

(3) $-1-2\sqrt{2}$ (4) $-1+\sqrt{2}$

ANSWERS AND SOLUTIONS

1. (1) **2.** (2) **3.** 16 **4.** (2) **5.** (2)

1. **(1)** Write $2 - x^2 = (1-x)(1+x) + 1$, so the integrand splits as $e^x\left(\sqrt{\frac{1+x}{1-x}} + \frac{1}{\sqrt{1+x}(1-x)^{3/2}}\right)$.

Let $g(x) = \sqrt{\frac{1+x}{1-x}}$. Then $g'(x) = \frac{1}{\sqrt{1+x}(1-x)^{3/2}}$.

Using $\int e^x(g+g')dx = e^x g + C$: $f(x) = e^x\sqrt{\frac{1+x}{1-x}} + C$.

$f(0) = 1 + C = 0 \Rightarrow C = -1$.

$f(1/2) = e^{1/2}\sqrt{\frac{3/2}{1/2}} - 1 = \sqrt{e}\cdot\sqrt{3} - 1 = \sqrt{3e} - 1$.

2. **(2)** Let $u = x^{1/6}$, so $x = u^6$ and $dx = 6u^5du$. Then $x^{2/3} = u^4$, $x^{1/2} = u^3$.

$f(x) = \int \frac{6u^5du}{u^4+2u^3} = \int \frac{6u^2du}{u+2}$.

By polynomial division: $\frac{u^2}{u+2} = u - 2 + \frac{4}{u+2}$.

$f(x) = 6\left(\frac{u^2}{2} - 2u + 4\ln|u+2|\right) + C = 3u^2 - 12u + 24\ln(u+2) + C$.

Substituting back: $f(x) = 3x^{1/3} - 12x^{1/6} + 24\ln(x^{1/6}+2) + C$.

From $f(0) = -26 + 24\ln(2)$: $24\ln(2) + C = -26 + 24\ln(2)$, so $C = -26$.

At $x = 1$: $f(1) = 3 - 12 + 24\ln(3) - 26 = -35 + 24\ln(3)$.

Thus $a = -35$, $b = 24$, and $a + b = -11$.

3. **(16)** $\int \sin^{-11/2} x \cos^{-5/2} x\, dx$.

Since $m + n = -8$ (negative even), substitute $t = \cot x$, $dx = \frac{-dt}{1+t^2}$.

Integral becomes $-\int \frac{(1+t^2)^3}{t^{5/2}}dt = -\int(t^{-5/2} + 3t^{-1/2} + 3t^{3/2} + t^{7/2})dt$.

$= \frac{2}{3}t^{-3/2} - 6t^{1/2} - \frac{6}{5}t^{5/2} - \frac{2}{9}t^{9/2} + C$.

Comparing: $\frac{p_1}{q_1} = \frac{2}{9}$, $\frac{p_2}{q_2} = \frac{6}{5}$, $\frac{p_3}{q_3} = \frac{6}{1}$, $\frac{p_4}{q_4} = \frac{2}{3}$.

$\frac{15p_1p_2p_3p_4}{q_1q_2q_3q_4} = \frac{15\cdot2\cdot6\cdot6\cdot2}{9\cdot5\cdot1\cdot3} = \frac{2160}{135} = 16$.

4. **(2)** $4x^2 + 8x + 3 = (2x+1)(2x+3)$ and $4x + 6 = 2(2x+3)$.

$I(x) = \frac{3}{4}\int \frac{du}{u^{3/2}\sqrt{u-2}}$ where $u = 2x + 3$.

Substituting $u = 2\sec^2\phi$: $I(x) = \frac{3}{4}\sin\phi + C = \frac{3}{4}\sqrt{\frac{2x+1}{2x+3}} + C$.

$I(0) = \frac{3}{4}\sqrt{\frac{1}{3}} + C = \frac{\sqrt{3}}{4} + C = \frac{\sqrt{3}}{4} + 20 \Rightarrow C = 20$.

$I\left(\frac{1}{2}\right) = \frac{3}{4}\sqrt{\frac{2}{4}} + 20 = \frac{3}{4\sqrt{2}} + 20 = \frac{3\sqrt{2}}{8} + 20$.

$a = 3, b = 8, c = 20$, $\gcd(3, 8) = 1$. $a + b + c = 31$.

5. **(2)** Substituting $u = \log t$ with $dt = e^u du$:

$f(t) = \int \frac{1-\sin u}{1-\cos u}e^u du$

Express as: $\frac{1-\sin u}{1-\cos u} = \frac{1}{2}\csc^2(u/2) - \cot(u/2)$

Using integration by parts and standard techniques:

$f(t) = e^{\log t}\cot(\frac{\log t}{2}) + C = t\cot(\frac{\log t}{2}) + C$

From $f(e^{\pi/2}) = -e^{\pi/2}$:

$e^{\pi/2}\cot(\pi/4) + C = -e^{\pi/2} \rightarrow e^{\pi/2} + C = -e^{\pi/2} \rightarrow C = -2e^{\pi/2}$

For $f(e^{\pi/4})$: $f(e^{\pi/4}) = e^{\pi/4}\cot(\pi/8) - 2e^{\pi/2}$

With $\cot(\pi/8) = 1 + \sqrt{2}$: $\alpha = -1 - \sqrt{2}$